

Article title: Deep Learning for Skin Cancer Image Analysis: A Systematic Literature Review

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Author names: Antônio Vinicius de M. Rodrigues, Roberta B. Oliveira, Flávio de Barros Vidal

Affiliation: Department of Computer Science, University of Brasilia, Brasilia, Brazil

E-mail address of the corresponding author: rodrigues.antonio@aluno.unb.br

Supplementary Information

This document contains the overview of the Supplementary Information and research datasets accompanying the manuscript “Deep Learning for Skin Cancer Image Analysis: A Systematic Literature Review”, submitted to the Computers in Biology and Medicine.

Files Overview

online_resource_1_prisma_checklist.pdf

Description: Completed PRISMA 2020 Checklist. It details the page numbers and sections where each recommended reporting item is addressed within the systematic review.

online_resource_2_excluded_reports.csv

Description: Excluded and Unretrieved Studies. A complete list of the 56 reports that were formally excluded during the full-text eligibility assessment, including the specific reasons for exclusion.

online_resource_3_llm_prompt.pdf

Description: LLM Data Extraction Prompt. The exact zero-shot prompt utilized for the programmatic deployment of the Large Language Model (LLM) during the automated data extraction phase, provided for methodological reproducibility.

online_resource_4_included_studies.csv

Description: Characteristics of Included Primary Studies. A comprehensive dataset outlining the extracted variables for all 254 included primary studies (e.g., computational tasks, datasets, base deep learning architectures, and primary evaluation metrics).

online_resource_5_quadas2_results.csv

Description: QUADAS-2 Risk of Bias Assessment. The domain-level results of the methodological quality and risk of bias assessment conducted for each of the 254 included studies using the adapted QUADAS-2 tool.

Figures Overview

figure_1.png

Description: Visual examples of the skin lesions (Benign lesion, Basal Cell Carcinoma, Squamous Cell Carcinoma, and Melanoma) sourced from the ISIC Archive.

figure_2.png

Description: High-level flowchart providing an overview of the fundamental computational tasks in skin cancer image analysis (Image Segmentation and Image Classification)

figure_3.png

Description: PRISMA 2020 Flow Diagram summarizing the systematic study selection process, automated filtering stages, and final corpus inclusion.

figure_4.png

Description: Polar coordinate chart illustrating the distribution of the retrieved records by publication year (angle) and their Impact Ratio percentile (radius) used during the quantitative assessment stage.

figure_5.png

Description: Choropleth maps demonstrating the geographical distribution of contributing countries based on author affiliations, before and after the filtering process.

figure_6.png

Description: Bar chart illustrating the overall distribution and prevalence of the dominant Deep Learning architectural families across the selected literature.